

TRIQX — AI Communication Assistant

Intelligent Smart Notification Hub & Headless Multi-Platform Response Engine

TARGET AUDIENCE

Executive & Engineering
Leadership

PLATFORM & BASELINE

Android 10+ (SDK 29–37) • MVVM
Clean

CORE TECHNOLOGIES

Kotlin, Jetpack Compose, Room,
Hilt, AI

PROJECT STATUS

Feature-Complete / Active API
Integration

01 Executive Summary

Triqx is an intelligent Android AI communication assistant engineered to solve **notification fatigue, fragmented app switching, and communication latency**. In modern professional workflows, users receive hundreds of incoming notifications daily across disjointed applications—WhatsApp, Microsoft Teams, Slack, Telegram, Gmail, and Outlook. Filtering urgent messages, switching apps to reply, and manually drafting responses consumes hours of cognitive focus each week.

Triqx bridges Android's native notification architecture with cloud and on-device AI intelligence to create a unified, centralized communications layer:

 Intelligent VIP Filtering

Filters out background noise, surfacing exclusively messages originating from whitelisted **Important Apps** or user-defined **Priority Contacts (VIPs)**.

 Zero-Latency AI Replies

Analyzes multi-turn conversation threads in real time to generate 1 to 5 context-aware reply suggestions instantly as messages arrive.

 1-Stage Headless Dispatch

Enables users to send replies directly from the **Android Notification Shade** or in-app hub—**without ever opening the native messaging apps**.

02 Business Value & Problem-Solution Matrix

INDUSTRY PROBLEM	HOW TRIQX SOLVES IT	MEASURABLE BUSINESS IMPACT
Notification Overload & Fatigue Critical stakeholder pings are buried beneath group chat noise and promotional alerts.	Rule-Based Dual Whitelisting: Ignores non-whitelisted traffic. Only authorized applications and VIP contacts trigger notification indexing and AI processing.	Zero Missed VIP Alerts: Eliminates distraction while providing 100% visibility for critical communications.
Context-Switching Drag Replying requires opening WhatsApp, then Slack, then Outlook—costing 15–30s per interaction.	Universal Headless Dispatch: Injects responses directly into Android RemoteInput slots or dispatches via official Cloud REST APIs in the background.	30–45 Mins Saved Daily: Removes application launching, splash screens, and navigation overhead entirely.
Repetitive Drafting Overhead Routine acknowledgments, confirmations, and scheduling queries are typed manually.	Contextual Smart Reply Engine: Considers previous chat history, contact identity, and subject headers to produce polished, 1-tap responses.	Near-Instant Response Times: Users can acknowledge, confirm, or defer communications with a single tap.
Generic "Robot" Responses Generic OS-level canned replies ("OK", "Thanks") sound robotic and lack contextual nuance.	3-Tier Multi-Layer Prompt Engine: Merges Global Persona directives, App-Specific Tones (e.g., Slack = Professional), and Conversation Overrides (e.g., Client = Formal).	Authentic, Professional Tone: Replies match exact personal tone, professional etiquette, and language requirements.

03 Core Product Capabilities ("What It Does")

Triqx bridges multiple messaging silos into a single, unified workflow. Rather than requiring the user to switch apps, Triqx acts as an ambient intelligent agent on the device.

Ingestion Multi-App Notification Parsing

Listens to system broadcasts and deconstructs incoming notifications across **WhatsApp, WhatsApp Business, Teams, Slack, Telegram, Gmail, and Outlook** into normalized conversation threads, isolating individual senders, email subjects, and thread context while eliminating synthetic noise.

Action Notification Drawer Assistant

Injects an interactive assistant card into Android's notification shade featuring a concise 2-line preview of incoming messages, AI suggested bullets, and **1 Send #1, 2 Send #2, 3 Send #3 action buttons** alongside an inline  **Edit RemoteInput** field.

Cloud Email Polymorphic Gmail & Outlook API

Employs official OAuth 2.0 (AppAuth) for Google and Microsoft Entra ID. Intelligently routes email replies through the Gmail REST API (RFC 2822 base64 MIME) or Microsoft Graph API (/v1.0/me/sendMail) based on the receiving account.

Intelligence 3-Tier Multi-Layer Prompt Merging

Allows granular persona management across three cascading layers: Global Settings (overall brevity/style), App Level (platform tone, e.g., Slack = Professional), and Conversation Level (individual contact language or specific rules).

Hub Unified Chat & Conversation Feed

Provides a centralized Google Messages / WhatsApp-style inbox where users can browse active threads across all apps, review full conversation history, inspect sender/receiver identifiers, and fine-tune per-thread AI reply counts (1–5) and prompts.

Auth Secure Mobile Phone OTP Flow

Includes a complete, production-ready onboarding experience featuring an international country code picker with search, SMS OTP verification, silent token refresh, and encrypted user profile management.

Key User Benefit: The user never needs to leave their active screen (e.g., reviewing a spreadsheet, reading documentation, or in a meeting) to read, review AI suggestions, or dispatch an authentic reply to a VIP client or colleague on WhatsApp, Slack, or Email.

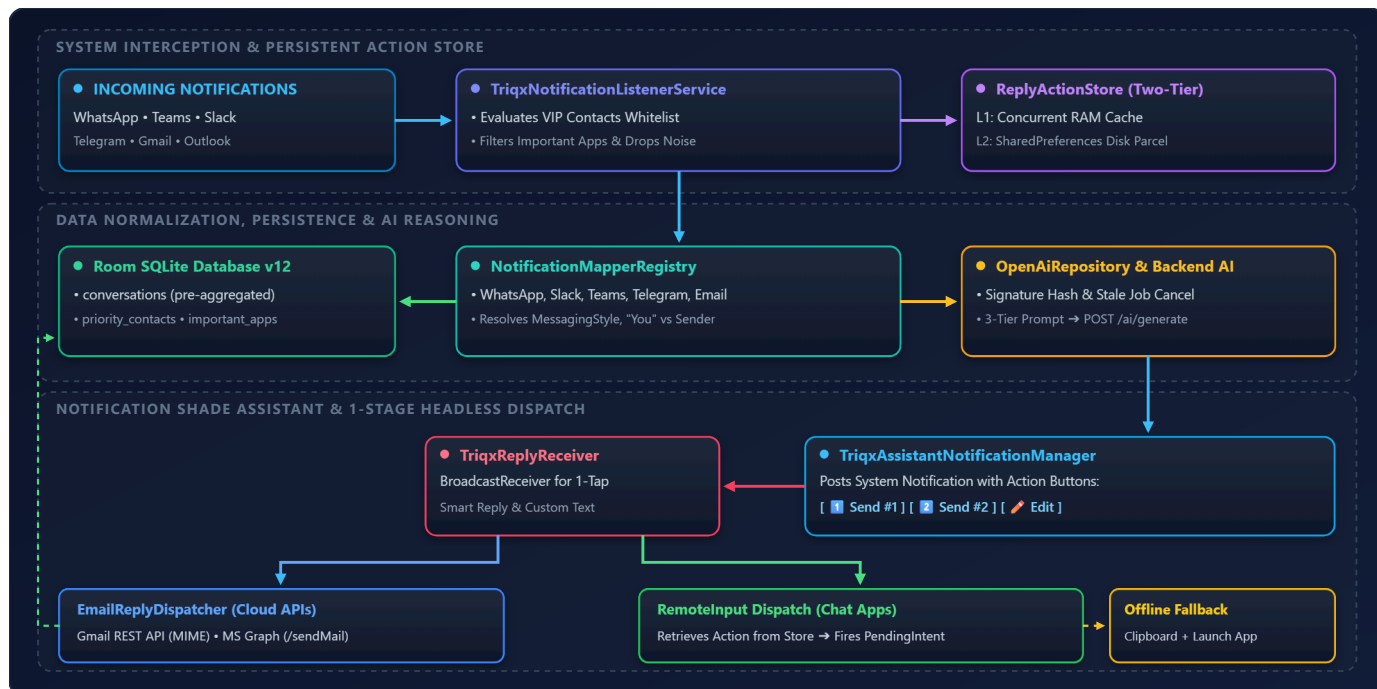
Granular VIP Contacts & Important Apps Management

Users maintain total authority over what reaches their priority feed:

- **Priority Contacts (VIPs):** Import directly from the Android address book with contact lookup key linking, or create manually with custom aliases, multiple phone numbers, and multiple email addresses.
- **Important Apps:** Select specific applications from all installed packages on the device. Each app can have its own dedicated reply style (e.g. Concise, Professional, Casual, Friendly) and custom instructions.
- **Strict Privacy Guard:** Notifications that match neither an important app nor a priority contact are completely ignored—never indexed, never saved, and never transmitted over the network.

04 Technical Architecture & Deep-Dive ("How It Does It")

Triqx is architected around **Clean Architecture**, **MVVM (Model-View-ViewModel)**, and **Unidirectional Data Flow (UDF)**. The system completely decouples notification capture, storage, AI orchestration, and dispatch into resilient, testable modules.



Core Engineering Innovations

1 Resilient Two-Tier Reply Action Store (ReplyActionStore)

The Challenge: Android's native `Notification.Action` and `PendingIntent` references are volatile objects destroyed by the OS when notifications are dismissed or the app process dies.

The Solution: Triqx serializes Android `Notification.Action` parcels into raw byte arrays, encodes them as Base64 strings, and stores them in persistent disk storage backed by an L1 in-memory cache. This allows Triqx to fire replies to conversations hours after the original notification was dismissed, and even across device reboots.

2 Modular Strategy Pattern Parsing Engine (NotificationMapperRegistry)

Dispatches notifications through specialized mappers (`WhatsAppNotificationMapper`, `TeamsNotificationMapper`, `SlackNotificationMapper`, `TelegramNotificationMapper`, `EmailNotificationMapper`, and `DefaultNotificationMapper`). Each mapper parses complex AndroidX `MessagingStyle` bundles, isolates group senders, extracts contact URIs, and strips platform-specific noise like WhatsApp's "👉 You got a reply".

3 In-Flight Job Debouncing & Deterministic Signature Caching

To prevent redundant API costs and UI flicker, OpenAiRepository hashes each incoming thread with a deterministic signature (sender|body|timestamp). If an identical message is received, cached replies are returned in 0ms. If a rapid follow-up message arrives while generation is in progress, the obsolete in-flight coroutine job is instantly cancelled via structured coroutine cancellation.

4 3-Tier Multi-Layer Prompt Merging Engine (resolveMergedPrompt)

Combines three independent prompt layers dynamically at runtime:

```
Final Prompt = [Global Prompt] ⊕ [App-Specific Prompt] ⊕ [Conversation-Specific Prompt]
```

Ensures that global corporate or user guidelines (e.g. "Keep under 20 words"), platform styles (e.g. "Professional & direct on Slack"), and relationship instructions (e.g. "Use formal German for Dr. Weber") are synthesized simultaneously.

5 Polymorphic Email Dispatch Subsystem (EmailReplyDispatcher)

Abstracts Google Gmail and Microsoft Outlook under a unified EmailService interface. When a reply is initiated, the system inspects the target account, acquires valid OAuth Bearer tokens (refreshing automatically if expired), and formats the appropriate cloud payload (RFC 2822 MIME for Gmail, JSON for Microsoft Graph) without launching third-party clients.

6 Graceful Multi-Stage Fallback Matrix

If a headless reply cannot be executed directly (e.g., an email account is not connected or a chat app revokes reply actions), Triqx initiates a 3-tier fallback sequence:

- **Tier 1:**
Headless RemoteInput / Cloud REST API (Preferred)
- **Tier 2:**
Deep-link ACTION_SENDTO Intent to pre-fill draft in target app
- **Tier 3:**
Auto-copy reply to clipboard and launch target application

05 Enterprise Security, Privacy & Reliability

Hardware-Backed Keystore Encryption

All session tokens (access, refresh) and sensitive user profile details are encrypted at rest via EncryptedSharedPreferences using **AES-256 GCM** encryption and **AES-256 SIV** keys backed by the hardware Android Keystore.

Strict Device-First Privacy Whitelisting

Notifications from non-whitelisted apps and non-VIP contacts are immediately discarded in memory. They are never transmitted over the network, processed by AI models, or saved in conversational storage.

Silent Background Token Refresh

If an AI or Auth API call encounters an HTTP 401 Unauthorized response, BackendAiService automatically intercepts the failure, executes a silent refresh via the refresh token, and retries the original request seamlessly.

Zero-Footprint Secure Sign-Out

When the user signs out, an exhaustive wipe is executed: all Room SQLite tables are cleared, email OAuth tokens are wiped, serialized reply actions are destroyed, active OS notifications are cancelled, and AI caches are flushed.

06 Technology Stack & Specifications

LAYER / CONCERN	TECHNOLOGY / LIBRARY	PURPOSE / ARCHITECTURAL ROLE
Language & SDK	Kotlin 100% (JVM 11) minSdk 29 (Android 10), targetSdk 37	Modern Android baseline with modern Coroutine concurrency, type safety, and Android 14+ platform parity.
UI Framework	Jetpack Compose (Material 3 Expressive) Navigation Compose, Coil, Chris Banes Haze	Declarative UI with fluid animations, adaptive layouts, and real-time glassmorphism / liquid blur aesthetics.
Architecture & Concurrency	MVVM + Clean Architecture Kotlin Coroutines, SupervisorJob, StateFlow	Decoupled, reactive state management with non-blocking I/O execution across background services.
Dependency Injection	Dagger Hilt + KSP	Compile-time dependency injection across singletons, background services, view models, and repositories.
Local Persistence	Android Room Database v12 SQLite, Room KTX, Custom TypeConverters	ACID-compliant storage for conversation history, VIP contacts, important apps, and diagnostic telemetry.
Cloud & Auth APIs	AppAuth-Android, Google Play Services Auth, OkHttp 3, Gson, Custom Phone OTP Service	Standards-compliant OAuth 2.0 token management for Gmail, Microsoft Graph, and Triqx backend microservices.

07 Current Milestones & Future Roadmap

✔ **Completed & Operational**

- **Cross-Platform Interception:** WhatsApp, Teams, Slack, Telegram, Gmail & Outlook.
- **Contextual AI Replies:** In-flight debouncing, 0ms cache hits & 3-tier prompt merging.
- **Persistent Reply Store:** Two-Tier RAM + Disk marshaling surviving OS reboots.
- **Notification Assistant:** 1-Tap numbered action buttons and inline RemoteInput editing.
- **Polymorphic Cloud Dispatch:** Background Gmail REST API & MS Graph API replies.

🚀 **Roadmap & Next Horizons**

- **On-Device LLM (Gemini Nano):** Local AI inference via Android AICore when offline.
- **Wear OS Companion:** 1-Tap smart reply triggers directly from wristwear.
- **Scheduled Dispatch:** Queue drafts to send automatically during business hours.
- **Enterprise Connectors:** Native integrations for Jira, Linear & Zendesk alerts.

08 Strategic Takeaways for Leadership

- **Solves High-Value Friction:** Eliminates the cognitive fatigue and lost productivity of jumping across 6 communication tools dozens of times daily.
- **Solves Android Platform Limits:** Overcomes volatile OS notification lifecycles via persistent parcel marshaling to disk, guaranteeing replies are never lost.
- **Production-Grade Architecture:** Engineered to the highest standards—Jetpack Compose, Clean Architecture, Hilt DI, AES-256 hardware encryption, and proactive token management.